

```
Open  p1.c
/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01 Save x

1#include<stdio.h>
2#include<unistd.h>
3#include<sys/wait.h>
4#include<stdlib.h>
5
6int main() {
7    pid_t pid;
8    char command[50];
9
10   printf("Enter command: ");
11   scanf("%49s", command);
12
13   printf("Parent process: \n");
14   printf("PID: %d\n", getpid());
15   printf("PPID: %d\n", getppid());
16
17   pid = fork();
18
19   if (pid < 0) {
20       printf("Fork() invocation failed\n");
21       exit(1);
22   } else if (pid == 0) {
23       printf("Child Process: %d\n", getpid());
24       printf("PID: %d\n", getpid());
25       printf("PPID: %d\n", getppid());
26
27       execlp(command, command, NULL);
28
29       printf("exec() failed\n");
30       exit(1);
31   } else {
32       printf("Parent Process\n");
33       wait(NULL);
34       printf("Parent Resumed\n");
35       printf("Child process finished\n");
36       exit(0);
37   }
38   return 0;
39 }
```

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de11@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ ./a.out
Enter command: ls
Parent process:
PID: 874
PPID: 307
Parent Process
Child Process: 875
PID: 875
PPID: 874
a.out p1.c
Parent Resumed
Child process finished
```

```
de11@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ uname -a
Linux DESKTOP-EFR7Q53 6.18.33.2-microsoft-standard-WSL2 #1 SMP PREEMPT_DYNAMIC Thu Jun 18 21:54:43 UTC 2
026 x86_64 GNU/Linux
de11@DESKTOP-EFR7Q53:/mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ lscpu
Architecture:          x86_64
CPU op-mode(s):        32-bit, 64-bit
Address sizes:          39 bits physical, 48 bits virtual
Byte Order:             Little Endian
CPU(s):                 12
On-line CPU(s) list:   0-11
Vendor ID:              GenuineIntel
Model name:             13th Gen Intel(R) Core(TM) i7-1355U
CPU family:             6
Model:                  186
Thread(s) per core:     2
Core(s) per socket:     6
Socket(s):              1
Stepping:               3
BogoMIPS:               5222.40
Flags:                  fpu vme de pse tsc msr pae mce cx8 apic sep mtrr pge mca cmov pat pse36 clf
                        lush mmx fxsr sse sse2 ss ht syscall nx pdpe1gb rdtscp lm constant_tsc rep_
                        good nopl xtopology tsc_reliable nonstop_tsc cpuid tsc_known_freq pni pclmu
                        lqdq vmx ssse3 fma cx16 pcid sse4_1 sse4_2 x2apic movbe popcnt tsc_deadline
                        _timer aes xsave avx f16c rdrand hypervisor lahf_lm abm 3dnowprefetch ssbd
                        ibrs ibpb stibp ibrs_enhanced tpr_shadow ept vpid ept_ad fsgsbase tsc_adjus
                        t bmi1 avx2 smep bmi2 erms invpcid rdseed adx smap clflushopt clwb sha_ni x
```

```

dell@DESKTOP-EFR7Q53: /mnt
Virtualization features:
  Virtualization:      VT-x
  Hypervisor vendor:   Microsoft
  Virtualization type: full
Caches (sum of all):
  L1d:                 288 KiB (6 instances)
  L1i:                 192 KiB (6 instances)
  L2:                  7.5 MiB (6 instances)
  L3:                  12 MiB (1 instance)
NUMA:
  NUMA node(s):        1
  NUMA node0 CPU(s):   0-11
Vulnerabilities:
  Gather data sampling: Not affected
  Ghostwrite:           Not affected
  Indirect target selection: Not affected
  Itlb multihit:        Not affected
  L1tf:                 Not affected
  Mds:                  Not affected
  Meltdown:             Not affected
  Mmio stale data:      Not affected
  Old microcode:        Not affected
  Reg file data sampling: Mitigation; Clear Register File
  Retbleed:             Mitigation; Enhanced IBRS
  Spec rstack overflow: Not affected
  Spec store bypass:    Mitigation; Speculative Store Bypass disabled via prctl
  Spectre v1:           Mitigation; usercopy/swapgs barriers and __user pointer sanitization

```

```

dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ lsblk
NAME MAJ:MIN RM  SIZE RO TYPE MOUNTPOINTS
sda   8:0      0 356.9M  1 disk
sdb   8:16     0 159.4M  1 disk
sdc   8:32     0    2G    0 disk [SWAP]
sdd   8:48     0    1T    0 disk /mnt/wslg/distro
/

dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ ps
  PID TTY          TIME CMD
  307 pts/0    00:00:00 bash
  917 pts/0    00:00:00 ps

dell@DESKTOP-EFR7Q53: /mnt/c/Users/DELL/OneDrive/Desktop/OS/2520030283_Practical/Practical-01$ top
top - 09:32:35 up 3 min, 1 user, load average: 0.07, 0.05, 0.01
Tasks: 35 total, 1 running, 34 sleeping, 0 stopped, 0 zombie
%Cpu(s): 0.0 us, 0.8 sy, 0.0 ni, 99.2 id, 0.0 wa, 0.0 hi, 0.0 si, 0.0 st
MiB Mem : 7784.6 total, 6066.9 free, 583.2 used, 1288.9 buff/cache
MiB Swap: 2048.0 total, 2048.0 free, 0.0 used. 7201.5 avail Mem

  PID USER      PR  NI   VIRT   RES   SHR S  %CPU  %MEM    TIME+  COMMAND
  927 dell      20   0   8144   5592   3520 R   9.1   0.1   0:00.01 top
    1 root      20   0  24108  15412 11624 S   0.0   0.2   0:00.76 systemd
    2 root      20   0   3180   2204   2072 S   0.0   0.0   0:00.01 init-systemd(Ub
    6 root      20   0   3180   2048   1940 S   0.0   0.0   0:00.00 init
   46 root      19  -1  42208  16812 15532 S   0.0   0.2   0:00.20 systemd-journal
   78 systemd+  20   0  22416  14480 11980 S   0.0   0.2   0:00.09 systemd-resolve
   89 root      20   0  35236  12304  9228 S   0.0   0.2   0:00.31 systemd-udev
  146 root      20   0   2888   1948   1820 S   0.0   0.0   0:00.05 chronyd-starter

```